

Robust Control Strategies applied to the PMSM Based on Programmable Digital Sensor Feedback

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Abstract: This paper presents advanced control strategies for Permanent Magnet Synchronous Machines (PMSMs). Due to their nonlinear characteristics, PMSMs require robust and adaptive control methods. The study first evaluates Field-Oriented Control (FOC) with proportional–integral (PI) controllers and identifies limitations caused by parameter variations and mechanical sensors. To address these issues, a robust control approach combined with Model Reference Adaptive System (MRAS) techniques and an observer for speed and position estimation is proposed. The results confirm the effectiveness of the MRAS-based strategy in ensuring accurate, sensorless control under low-speed and challenging conditions.

Keywords: PMSM, Robust and Adaptive Control, FOC, PI, Observer, MRAS

1. INTRODUCTION

Mechanical position and speed sensors, such as encoders and resolvers, are increasingly eliminated in modern drive systems to enhance reliability and reduce cost [1], [2], [16] and [17]. Sensorless control, based only on stator current and voltage measurements, removes mechanical components but faces challenges at low and zero speeds [2]. Saliency tracking methods including high frequency injection [3], [4] and transient excitation [5], [6] offer good low-speed performance but introduce additional losses and noise. Observer based approaches, such as flux observers [7], [8], SMO [9], and EKF [10], [11], provide simpler implementation yet suffer from chattering issues. This work analyzes machine observability and proposes a PI-MRAS based observer for accurate sensorless estimation of speed and position, ensuring reliable operation even in low-speed conditions.

2. SPEED ESTIMATION USING THE MRAS TECHNIQUE (MRAS OBSERVER)

MRAS adjusts system parameters by minimizing the error between the adaptive model (AM) and the reference model (RM).

Principle of the MRAS Technique

The Model Reference Adaptive System (MRAS) is an indirect speed estimation method that uses measured stator voltages (V_d , V_q) and currents (I_d , I_q) [12], [13]. It

employs two models: a Reference Model, which calculates stator current vectors in the d–q frame, and an Adaptive Model, which estimates them from measured signals. The goal is to minimize the error between both models, ensuring accurate and convergent speed estimation. A PID controller is typically used to adapt the system by aligning the adaptive model with the reference model.

Figure 1 illustrates the speed estimation method using MRAS.

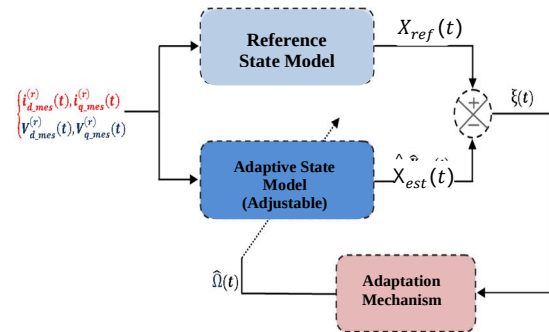


Fig. 1 Representative block diagram of the MRAS observer

Formulation of the MRAS Observer

Consider the design of the two stator current estimators based on the MSAP state-space model, using measured stator voltages and currents [12], [13].

Using the MSAP state-space model, the state equations for the two current components in the rotor reference frame can be expressed as follows:

demonstrates that the observer can reliably fulfil its observation function. Figure 7 shows the superposition of the estimated and measured positions during this test.

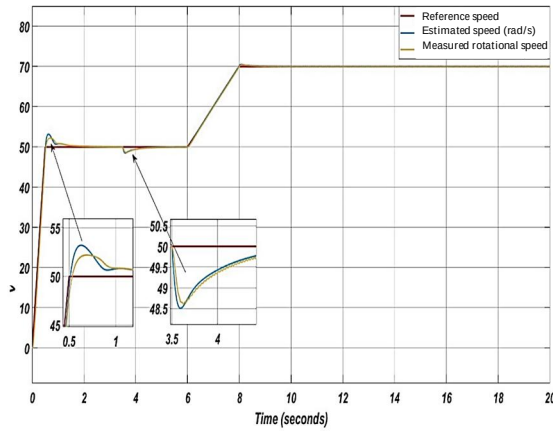


Fig. 3 Sensorless speed control using the MRAS method.

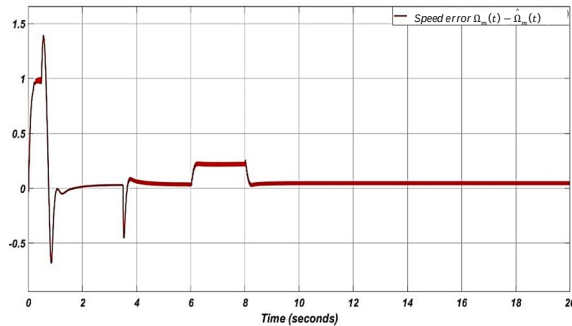


Fig. 4 Difference between the actual speed and the estimated speed

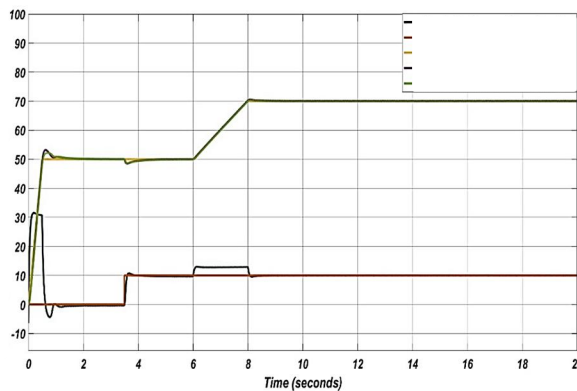


Fig. 5 Rotor speed response with load torque disturbance at $t = 3.5$ s, $T_l = 10$ N·m (reference).

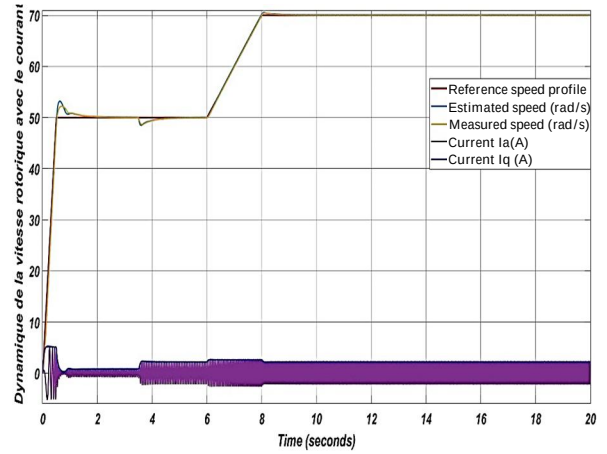


Fig. 6 Current profile in sensorless control as a function of the speed response.

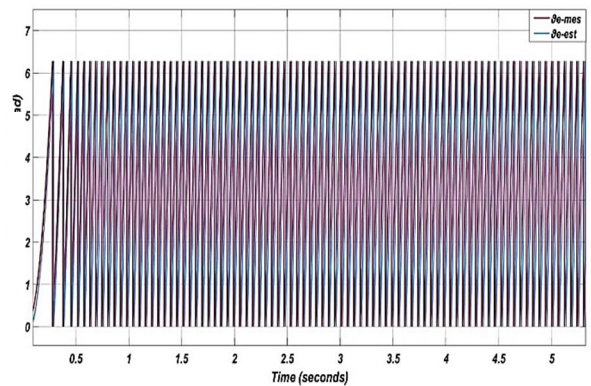


Fig. 7 Real and estimated machine positions

4. CONCLUSION

This study analyzed the observability of the machine as a foundation for sensorless control and proposed an observer-based approach combining PI vector control with the MRAS technique. The results demonstrate that MRAS enables accurate speed and position estimation without mechanical sensors, ensuring robust performance even under low-speed and demanding operating conditions.

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